Coursework Title:	HCI Essay
Course Name:	Human-Computer Interaction
Course Code:	DECO2500/7250
Weighting:	20%
Course Coordinator:	Dr Nell Baghaei
Contact:	Room: 78-304
Hand-in Date:	14th October 2022 by 4:00pm (AEST)
Hand-in Method:	Blackboard

You are required to choose **either** Assignment 4A **or** Assignment 4B to work on. You are **not** required to complete both.

This is an **individual** piece of work. You are not allowed to work with anyone else during the completion of this assignment.

You must submit your work via Blackboard on or before the due date/time.

Learning Objective Being Assessed

2 Relate cognitive, social, behavioural, and emotional theories to the design of effective human-system interaction.

6 Apply HCI knowledge, in terms of system accessibility, within appropriate user-interface design for a broad range of user groups.

Assignment 4A:

1. Introduction

You are required to write an essay about an HCI or UX concept or method of your choice. The essay must incorporate an explanation of the concept or method, research, a case study of that concept or method, and a critical analysis of the concept or method outlining issues such as (but not necessarily including or limited to) limitations of its application to emerging technology or problems arising in the application of an evaluation method.

2. Details of the Task

Throughout the course you will have encountered many concepts related to Human-Computer Interaction and User Experience. You will have also encountered and used many methods that are used to either enquire into or evaluate a phenomenon. Earlier in the semester, you encountered concepts such as conceptual models for a system and different issues related to human cognition. As you progress through the Interaction Design process, you will use a number of different methods to evaluate your prototypes.

Your experience from the group project has provided you with an opportunity to gather insights into the application of concepts or use of an HCI method within the context of your project.

Through this assignment, you will be presenting an individual reflective essay that encompasses both background to the concept or method that you have chosen, and what you have learned through its application in your project.

You are required to describe in your essay:

a) An explanation of your chosen concept or method

b) Where applicable, this should incorporate the development of that concept or method, theories on which it is based, and how that concept or method is being applied in current practice, i.e., in research and/or industry.

c) A case study of the application of that concept or method based on the work that has been completed in your group work. You will need to use evidence drawn from *your own* HCI process as data for the analysis. Your case study might include, for instance:

- Explanation of a documented workshop in which you applied your chosen method to your project, where you can show how the method was carried out and what value it delivered for your team at that stage of the process
- A demonstration (e.g., through visible changes over prototype iterations, or iterations of your conceptual design), of how the use of a particular HCI concept influenced or altered your project and/or your understanding of the problem or context

d) A critical analysis of that concept or method based on your experience of its application in the project. This critical analysis should include considerations of limitations or practical issues arising from its application or use. Examples can include (but are not limited to):

- Limitations in terms of the use and nature of current or emerging interactive systems.
- Potential practical problems in the use of an evaluation method given the form of the technology, the particular user community or the context of use.
- How a concept or method might need to change or evolve to take into account current approaches to, for example, defining User Experience goals or the nature of Design Principles.

The body of your essay should be 1500 words (excluding reference list). You should aim to be close to this word limit (e.g. within ±10%). **The content of your essay must be completely independent from any work undertaken within your group (i.e. do not directly copy work from your group assignments into your essay).**

You must appropriately reference all resources in your essay and include a reference list. IEEE referencing style is recommended but not required. It is your choice what referencing style you ultimately use. Whichever style you choose, ensure you are consistent throughout your document. You are advised to use a word processor (such as Microsoft Word or OpenOffice/LibreOffice) that supports spelling and grammar checking to prepare your essay.

3. Grading Scheme/Assessment Criteria*

Assessment Criteria		% weighting for criteria
Essay	Explanation of Content	20
	Research and Literature	30

	Case Study	20
	Critical Analysis	20
	Structure, Writing Style, and Presentation	10

**please refer to the rubric on pages 6 – 7 for a detailed breakdown of each criteria*

Assignment 4B:

Choose an interface (a website, or a mobile application or a game or a AR/VR application) and evaluate it using Nielsen's ten usability heuristics (See Week 6 slides). Your report should have an *Introduction* section (in which you introduce the interface), *Critical Evaluation* section (divided into 10 subsections based on 10 usability heuristics) and *Conclusions and Areas of Improvement.* For each heuristic, explain why you think the interface of your choice meets or doesn't meet the heuristic and justify your explanation. Use of screenshots are highly recommended.

Examples you can choose from include: Gmail (web or mobile version), Blackboard, Zoom, Microsoft Teams, UQ website, Commonwealth Bank, ANZ Bank, An Australian govt website or mobile app, PeerWise, YouTube, Amazon, Facebook/Meta, Instagram, Bing, Netflix, TikTok, Snapchat, Telegram, WhatsApp, WeChat, Qantas, Alibaba, a mobile game of your choice, a VR/AR game, a VR/AR app, etc.

The body of your report should be 1500 words (excluding screenshots and reference list). You should aim to be close to this word limit (e.g. within ±10%).

Submit the summary through Blackboard by the due date (14 October).

References (if you have any): Please use APA 6th referencing style: https://guides.library.uq.edu.au/referencing/apa6

Grading Scheme/Assessment Criteria*

Assessment Criteria		% weighting for criteria
Report	Critical Evaluation using ten usability heuristics	60
	Introduction/Conclusions/Areas of Improvement	20
	Structure, Writing Style, and Presentation	20

**please refer to the rubric on page 8 for a detailed breakdown of each criteria*

What You Should Hand In For Assignment 4A OR 4B

You should submit a **single electronic file** via Blackboard on or before the due date/time.

Your submission must be a single word-processed document that contains your essay, including reference list (e.g. Microsoft Word (.doc/.docx) or Pages (.pages) formats only).

No other forms of submission are allowed without prior written permission from the course coordinator.

Guidelines

- Correctly reference any resources that you use.

Resources Required

You should make use of these specific tools and resources:

- Lecture/Studio Materials
- The Internet
- Course-specific textbooks
- UQ Resources, such as:
  - https://my.uq.edu.au/information-and-services/student-support/study-skills/assignment-writing
  - https://my.uq.edu.au/information-and-services/student-support/study-skills/assignment-writing/assignment-types
  - https://my.uq.edu.au/information-and-services/student-support/study-skills
  - https://web.library.uq.edu.au/research-tools-techniques/google-scholar
  - https://guides.library.uq.edu.au/referencing
- Published academic papers on relevant methods or concepts, such as (but not limited to just) those below, which may give you an idea about writing style, critical reflections on methods and assessments of their value to particular projects:
  - Blomkvist, J., & Arvola, M. (2014). *Pausing or not? : Examining the Service Walkthrough Technique*. 171–176.
  - Tan, W., Liu, D., & Bishu, R. (2009). Web evaluation: Heuristic evaluation vs. user testing. *International Journal of Industrial Ergonomics*, *39*(4), 621–627. https://doi.org/10.1016/j.ergon.2008.02.012

Late Submission/Extenuating Circumstances

The submission of progressive assessment material on the due date as set out in the Electronic Course Profile (ECP) is the sole responsibility of the student. Students should not leave assignment preparation until the last minute and must plan their workloads to meet advertised or notified deadlines. It is your responsibility to manage your time effectively.

Where the assessment item is submitted after the deadline, without an approved extension, a late penalty will apply. The late penalty shall be one grade will be deducted from the grade for the assessment item per calendar day (or part thereof). A day is considered to be a 24-hour block from the assessment item due time. E.g. if you originally received a Credit but your submission was late and within 24 hours you will lose one grade and will be awarded a Pass. If you are over 24 hours late, you will continue to lose one grade in this way per 24-hour block until the end of the scale is reached. Negative grades will not be awarded.

However, if there are medical or exceptional circumstances that will affect your ability to complete an assessment by the due date, then you can apply for an extension. Extensions to

all other assessment items must be requested via my.UQ. You can find instructions on how to submit your request online.

An extension application granted on medical grounds will not be approved for any more than the number of calendar days the medical certificate indicates you were unfit for study. Students who are ill for more than 14 days should consider applying for withdrawal without academic penalty.

Requests must be made *at least* 48 hours prior to the submission deadline, unless the medical or other circumstances are such that you could not reasonably be expected to have applied by then. Requests for extensions which are received on or after the due date may not be considered. Please refer to the ECP for full details about the University's policies and guidelines regarding assessment.

Academic Misconduct

The University defines Academic Misconduct as involving 'a range of unethical behaviours that are designed to give a student an unfair and unearned advantage over their peers'.

The University takes Academic Misconduct very seriously and any suspected cases will be investigated through the University's standard policy. If you are found guilty, you may be expelled from the University with no award.

It is the responsibility of the student to ensure that you understand what constitutes Academic Misconduct and to ensure that you do not break the rules. If you are unclear about what is required, please ask.

It is also the responsibility of the student to take reasonable precautions to guard against unauthorised access by others to their work, however stored in whatever format, both before and after assessment.

For more information, you are directed to the following University web pages:

- Information regarding ***ITEE Student Misconduct (Including Plagiarism)***
  - https://www.itee.uq.edu.au/itee-student-misconduct-including-plagiarism
- Information regarding ***Academic Integrity and Misconduct***:
  - https://my.uq.edu.au/information-and-services/manage-my-program/student-integrity-and-conduct/academic-integrity-and-student-conduct
  - http://ppl.app.uq.edu.au/content/3.60.04-student-integrity-and-misconduct
- Information on ***Student Services:***
  - https://www.uq.edu.au/student-services/
- Information regarding ***Referencing***:
  - https://web.library.uq.edu.au/research-tools-techniques/digital-essentials/write-cite-submit?p=2#2
  - https://web.library.uq.edu.au/research-tools-techniques/referencing-style-guides
- Information regarding the ***Student Charter***:
  - https://ppl.app.uq.edu.au/content/3.60.01-student-charter
- Information regarding ***Plagiarism***:
  - https://web.library.uq.edu.au/node/4221/1#1

Essay Grading Criteria for Assignment 4A

Criteria/ Weighting	High Distinction *Demonstrated evidence of **exceptional** achievement in relation to assessment criteria*	Distinction *Demonstrated evidence of **advanced** achievement in relation to assessment criteria*	Credit *Demonstrated evidence of **proficient** achievement in relation to assessment criteria*	Pass *Demonstrated evidence of **functional** achievement in relation to assessment criteria*	Fail ***Minimal to Absence** of evidence of achievement in relation to assessment criteria*	X (None) *No assessable work has been received*
Explanation of Content (20%)	Explanation of the concept/method demonstrates an **exceptional in-depth** and **thorough** understanding of the concept/method.	Explanation of the concept/method demonstrates an **advanced** and **reasonable** understanding of the concept/method.	Explanation of the concept/method demonstrates a **competent** and **satisfactory** understanding of the concept/method.	Explanation of the concept/method demonstrates an **acceptable** but **limited** understanding of the concept/method.	Explanation of the concept/method demonstrates a **deficient** understanding or misunderstanding of the concept/method.	This component is missing
Research and Literature (30%)	Research has demonstrated **exceptional** breadth through incorporation of a number of relevant academic papers to demonstrate **mastery** of the content in the area of HCI. Your application of this research captures **superior** detail to illustrate a **thorough** understanding of HCI.	Research has demonstrated an **advanced** level of breadth through incorporation of a number of relevant academic papers to demonstrate **substantial** knowledge of fundamental concepts in the area of HCI. Your application of this research captures a **suitable** level of detail to illustrate a **reasonable** understanding of HCI.	Research has demonstrated a **competent** level of breadth to demonstrate **good** knowledge of fundamental concepts in the area of HCI. Incorporation of a number of relevant academic papers may be lacking. Your application of this research captures **adequate** detail to illustrate a **satisfactory** understanding of HCI.	Research has demonstrated an **acceptable** level of breadth to demonstrate **adequate** knowledge of fundamental concepts in the area of HCI. Incorporation of a number of relevant academic papers may be very limited. Your application of this research captures **minimal** detail and illustrates a **limited** understanding of HCI.	Research has demonstrated a **severe lack** of detail and illustrates **deficiencies** or **no demonstratable evidence** of understanding of the fundamental concepts in the area of HCI. Very limited to no academic papers have been identified.	
Case Study (20%)	Case study **clearly** demonstrates evidence drawn from your own HCI process.	Case study **suitably** demonstrates evidence drawn from your own HCI process.	Case study **adequately** demonstrates evidence drawn from your own HCI process.	Case study **minimally** demonstrates evidence drawn from your own HCI process.	Case study demonstrates **very little** detail of evidence drawn from your own HCI process.	

Criteria/ Weighting	High Distinction *Demonstrated evidence of **exceptional** achievement in relation to assessment criteria*	Distinction *Demonstrated evidence of **advanced** achievement in relation to assessment criteria*	Credit *Demonstrated evidence of **proficient** achievement in relation to assessment criteria*	Pass *Demonstrated evidence of **functional** achievement in relation to assessment criteria*	Fail ***Minimal to Absence** of evidence of achievement in relation to assessment criteria*	X (None) *No assessable work has been received*
Critical Analysis (20%)	Critical analysis of the concept/method presents **significant** and **sophisticated** insights through an **in-depth,** reasoned, analytical, and constructive argument. A **thorough** description of relevant issues have been identified with respect to the concept/method that are based on real experience.	Critical analysis of the concept/method presents **valuable** insights through an **advanced,** reasoned, analytical, and constructive argument. A **reasonable** description of relevant issues has been identified with respect to the concept/method that are based on real experience.	Critical analysis of the concept/method presents **coherent** insights through a **competent** argument. A **satisfactory** description of relevant issues has been identified with respect to the concept/method that are based on real experience.	Critical analysis of the concept/method presents **limited** insights through an **acceptable** argument. A **limited** description of relevant issues has been identified with respect to the concept/method that are mostly based on real experience.	Critical analysis of the concept/method presents **deficient** insights through a **poor or invalid** argument. A **deficient** description or **no demonstratable evidence** of relevant issues have been identified with respect to the concept/method.	This component is missing
Structure, Writing Style, and Presentation (10%)	The essay has been presented to an **exceptional** standard. It has been organised **very well** in a logical manner, adheres to the word limit and is **very easy** to follow. It communicates at an **expert** postgraduate level with minimal to no spelling/grammar errors.	The essay has been presented to an **advanced** standard. It has been organised **quite well,** the word limit has been reasonably adhered to and is **easy** to follow. It communicates at a **professional** postgraduate level. May contain limited spelling/grammar errors.	The essay has been presented to a **competent** standard. It has been organised **well,** the word limit may have been mostly adhered to and is **fairly easy** to follow. It communicates at an **effective** postgraduate level. May contain some spelling/ grammar errors.	The essay has been presented to an **acceptable** standard. It has been organised **adequately** but may be hard to follow. The word limit may not have been adhered to well. It communicates at an **appropriate** postgraduate level. May contain numerous spelling/grammar errors.	The essay **fails** to meet the standard at a postgraduate level. It is **disorganised** and communicates information or ideas in ways that are frequently incomplete, confusing, or inappropriate, with little consideration for the word limit. Written expression is **disjointed,** and the essay is badly structured with constant errors in spelling/ grammar.	

Essay Grading Criteria for Assignment 4B

CRITERIA/WEIGHTING	FAIL	PASS	CREDIT	DISTINCTION	HIGH DISTINCTION
CRITICAL EVALUATION USING NIELSEN'S TEN USABILITY HEURISTICS *Screenshots of the interface (60%)*	Minimal or no critical evaluation of the website using the usability heuristics. Insufficient explanation. The student has no or limited understanding of the underlying concepts.	Adequate level of critical evaluation using the usability heuristics. The submission demonstrates at least a basic understanding of the underlying concepts.	Appropriate level of critical evaluation. The submission demonstrates a sound understanding of the underlying concepts	Good level of critical evaluation. The student has demonstrated a clear level of understanding of the underlying concepts.	Excellent level of critical and insightful evaluation. The student has demonstrated a high level of understanding of the underlying concepts.
INTRODUCTION, CONCLUSION & AREAS OF IMPROVEMENT *References (optional) (20%)*	No introduction and/or conclusion. Limited awareness of audience or purpose of the website. Paragraphs lack development and coherence.	Adequate awareness of audience or purpose of the website. Basic introduction and conclusions and adequate suggestions for future improvements are provided.	Appropriate awareness of audience or purpose of the website. Sound introduction and conclusions and appropriate suggestions for future improvements are provided.	Good awareness of audience or purpose of the website. Clear introduction and conclusions and good suggestions for future improvements are provided.	Excellent awareness of audience or purpose of the website. Insightful introduction and conclusions and insightful suggestions for future improvements are provided.
STRUCTURE, WRITING STYLE & PRESENTATION *(20%)*	Incorrect and/or ineffective language. Poor presentation of the report, and/or structure. The submission demonstrates a limited of flow of concepts, and use of images/ text/materials.	Adequate use of language, structure and style, with some errors. The submission demonstrates a basic flow of concepts, and use of images/ text/materials.	Appropriate use of language, structure and style. The submission demonstrates a sound flow of concepts, and use of images/ text/materials.	Good use of language, structure and style. The submission demonstrates a clear flow of concepts, and use of images/ text/materials.	Excellent level of clarity in style and structure, and effective communication throughout. The submission demonstrates a coherent flow of concepts, and use of images/ text/materials.